THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 4: **Exploring Magnets** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which sentence is correct? 1

(a) a magnet can attract iron without touching it
(b) a magnet attracts plastic
(c) a magnet must touch iron
(d) a magnet attracts wood

2. Which instrument depends on the Earth's magnetic field? 1

(a) compass **(b)** clock **(c)** ruler **(d)** balance

3. A magnetised needle has one end that is attracted by the north pole of a bar magnet. That end is a: 1

(a) south pole **(b)** demagnetised **(c)** north pole **(d)** neutral

4. A magnet picks up pins through a glass sheet because: 1

(a) glass is thin **(b)** magnetic force passes through glass
(c) pins are light **(d)** glass is magnetic

5. If the north end of a compass needle is pulled towards the geographic north, then the needle's other end points: 1

(a) south **(b)** east **(c)** north **(d)** west

6. A bar magnet is rolled in iron filings. Which part picks up the fewest filings? 1

(a) the north pole **(b)** the south pole
(c) the middle **(d)** both ends equally

7. What would be the effect on a magnet's strength if it is dropped repeatedly? 1

(a) it weakens **(b)** it doubles **(c)** no change **(d)** it strengthens

8. When two magnets are put together with unlike poles, the combined magnet: 1

(a) has no poles **(b)** is longer and still has two poles
(c) has four poles **(d)** loses its poles

9. Which part of a bar magnet picks up the most iron pins? 1

(a) the flat sides near the middle **(b)** the middle
(c) the poles **(d)** the label

10. A magnet is placed in a pile of iron filings and lifted. Where are most filings seen? 1

(a) nowhere **(b)** in the middle **(c)** near the poles **(d)** uniformly

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does an iron nail kept near a magnet for a long time become weakly magnetic? 2

12. Why does iron rust but still get attracted by a magnet? 2

13. Why do magnets of strong strength stick together? 2

14. Why are electronic gadgets kept away from magnets? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe an experiment to show that a magnet's force acts through water. 3

16. Why are cranes with powerful magnets used in scrapyards? 3

17. How can one prepare a magnet that keeps its magnetism for long? 3

OR

How would you distinguish a magnet from a piece of iron? 3

18. Explain how magnetic force differs from a push or pull by contact. 3

OR

Differentiate natural and artificial magnets. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A student wants to make a compass. Describe the materials, steps and how to test it. 5

OR

A bar magnet is cut in half. Write the result and test it with filings. 5

20. Explain why heating, hammering and dropping weakens a magnet. Suggest a way to prevent it. 5

OR

Why are credit cards and phones damaged near magnets? Suggest safety rules. 5

END OF QUESTION PAPER — Total: 40 Marks

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